

Ammonia-fired chemically recuperated gas turbine: Thermodynamic analysis of cycle and recuperation system

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ABSTRACT

Ammonia is one of the prospective alternatives to hydrocarbon fuels. Currently, there are projects for developing of ammonia fired gas turbines up to 40 MW. The addition of thermochemical exhaust gas heat recuperation systems to ammonia-fired gas turbines could be a promising way to increase their efficiency. In this paper, the concept of an ammonia-fired chemically recuperated gas turbine (CRGT) is thermodynamically analyzed. Gas turbine with thermochemical recuperation by ammonia decomposition is analyzed via Aspen HYSYS for a wide range of operating parameters: turbine inlet temperature of 700–1300 °C, the pressure of 6–21 bar. The thermochemical exhaust heat recuperation system is recovering the exhaust heat in a reformer (for the endothermic reaction of ammonia decomposition), a heater (to preheat ammonia to the temperature of the decomposition reaction), and a regasifier (for regasification of liquid ammonia). The thermochemical exhaust heat recuperation system makes it possible to recover up to 43% of exhaust heat. The maximum efficiency of CRGT is observed at 9 bar for $T_{in} = 700$ °C; 12 bar at $T_{in} = 800$ °C; 15 bar at $T_{in} = 900$ °C; 18 bar at $T_{in} = 1000$ °C. In the temperature range above $T_{in} > 1000$ °C, when the pressure rises above 15 bar, the efficiency does not increase significantly.

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1. Introduction

The growth of the world economy in the 20th and early 21st centuries is associated with an increase in the consumption of hydrocarbon fuels. According to a report by the International Energy Agency, the consumption of hydrocarbon fuels in the global energy balance is increasing from year to year [1]. The increase in the use of hydrocarbon fuels leads to an increase in CO₂ emissions. It is recognized by the world scientific community that an increase in CO₂ emission leads to an increase in average world temperature. Therefore, the leading countries have adopted programs for the transition to carbon-free energy. The European Union plans to transfer energy with zero CO₂ emissions by 2050 [2]. China plans to abandon coal-fired power plants, which are one of the main sources of CO₂ emission [3]. Japan and the United States also presented their programs for decarbonization in the next dozen years [4].

One of the ways in the future carbon-free energy industry is the use of ammonia as a fuel. Ammonia is a gaseous carbon-free fuel

with a lower heating value of 22.5 MJ/kg. Ammonia can be used as a fuel in existing thermal power plants after their minor modernization. Currently, there are a lot of publications on the use of ammonia and ammonia-rich gas in internal combustion engines. Zamfirescu and Dincer described the potential of ammonia as a sustainable fuel [5]. Dimitriou and Javadi published a comprehensive review of the use of ammonia for compression ignition engines where showed the perspectives for ammonia using in marine, power generation, and possibly heavy-duty applications [6]. Yapicioğlu and Dincer presented a review on the use of clean ammonia as a potential fuel for power generators where discussed the perspectives of ammonia using as carbon-free fuel [7].

In March 2021, engineers from Japan announced that 40 MW gas turbine fired by 100% ammonia is on the final stage of developing [8]. Such a gas turbine will facilitate the decarbonization of small and medium-sized power plants for industrial applications. In such a gas turbine plant, in the first stage, ammonia is regasified with hot water at a pressure of 15 bar and then gaseous ammonia is used as fuel. In addition, there are a set of publications on the use of ammonia as a fuel for micro-gas turbines [9,10].

In addition to the advantages of ammonia as a fuel, it should be noted that ammonia is a good carrier of hydrogen. There are a set of

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publications on the use of ammonia as a hydrogen carrier [11–13]. In these publications, ammonia is initially used to produce a hydrogen-rich gas by endothermic reaction of ammonia decomposition. To produce a hydrogen-rich gas, external heat is used for ammonia decomposition.

The use of external heat from waste heat to produce hydrogen-rich fuels from ammonia is consistent with the concept of thermochemical exhaust heat recuperation [14,15]. The ammonia decomposition reaction is endothermic, therefore, for its implementation, an external heat supply is required [16]. Exhaust heat after fuel-consuming equipment can act as such external heat. Mardani et al. considered the low carbon vehicles fired by ammonia decomposition products produced with the help of exhaust gas heat [17]. Sittichompoo et al. investigated an exhaust heat recuperation system based on catalytic ammonia decomposition for hydrogen production in low carbon clean vehicles [18].

The combination of gas turbines fired with ammonia and thermochemical recuperation has great prospects. The scheme of thermochemical exhaust heat recuperation in a gas turbine unit was first proposed by Olmsted and Grimes [19]. They suggested using methane reforming before combustion to recuperate exhaust heat. Further, thermochemical recuperation in a gas turbine was widely discussed in the scientific literature, and a gas turbine with a thermochemical exhaust heat recuperation began to be called a chemically recuperated gas turbine (CRGT) [20–22].

Various schemes of chemically recuperated gas turbines were proposed for small turbines [20,23] and microturbines [24]. Kesser et al. performed the thermodynamic analysis of a gas turbine power plant with thermochemical recuperation via steam methane reforming [25]. The authors determined that the CRGT with a compressor pressure ratio of 30 have a thermal efficiency of about 47.0% while for a pressure ratio of 15, the thermal efficiency is about 47.3%. Moreover, one of the perspective ways of using of exhaust heat is integration of high temperature solid oxide fuel cell coupled with recuperative gas turbine [26]. Choudhary showed that the maximum efficiency of 73.46% can be achieved for recuperative gas turbine with solid oxide fuel cells [27].

Verkhivker and Kravchenko carried out the thermodynamic analysis of a combined cycle plant with thermochemical recuperation based on steam methane reforming [28]. The authors proposed using steam for reforming from a high-pressure cylinder of a steam turbine. Verkhivker and Kravchenko calculated the energy efficiency of a combined cycle plant with thermochemical recuperation has reached up to 90%.

Sadeghi et al. presented the results of thermoeconomic comparison and multi-objective optimization of combined gas turbine and organic Rankine cycle power generation system with thermochemical recuperation based on steam methane reforming and thermophysical recuperation [29].

It is to be expected that the use of the principles of thermochemical recuperation in ammonia-fired gas turbines will increase their effectiveness. In this case, exhaust heat after ammonia fire GT will be used to occur the reaction of ammonia decomposition as well as ammonia regasification and ammonia preheating. The thermodynamics of ammonia decomposition reaction are depending on operating parameters such as temperature and pressure. In turn, the efficiency of the gas turbine cycle is also depending on temperature and pressure. Therefore, the study of ammonia-fired chemically recuperated gas turbines is of interest.

In this paper, we performed the thermodynamic analysis of a chemically recuperated gas turbine fired with 100% of ammonia to determine the effect of operating parameters such as temperature and pressure on the efficiency. Also, we compared the efficiency of CRGT fired with ammonia and gas turbine without a recuperation system. In addition, the efficiency of the thermochemical exhaust

heat recuperation system as a function of temperature and pressure was determined and analyzed.

2. Methodology

2.1. Thermochemical recuperation concept

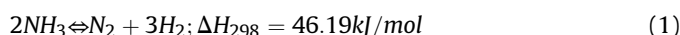
The main concept of the chemically recuperated gas turbine is using the hot exhaust gases after the gas turbine for a thermochemical transformation (endothermic decomposition) of ammonia to produce hydrogen-rich gas. Further, a hydrogen-rich fuel is used for heat generation in a combustion chamber. Since the reaction of ammonia decomposition is strongly endothermic, external heat has to be supplied. In CRGT, the source of external heat is exhaust heat after the gas turbine.

A CRGT concept can be explained by a schematic diagram from Fig. 1, which shows the schematic diagram of CRGT fired with ammonia. CRGT is can be virtually divided into subsystems: power generation subsystem and thermochemical recuperation subsystem. The power generation subsystem is classical. In the thermochemical recuperation subsystem, the exhaust gases after the gas turbine are passed through a reformer, a heater, and a regasifier. In this case, the heat recuperation occurs by the following stages:

- in a reformer for the endothermic reaction of ammonia decomposition;
- in a heater to preheat gaseous ammonia to the temperature of decomposition reaction;
- in a regasifier to evaporate liquid ammonia.

As it can be seen from Fig. 1, the heat generation from ammonia in CRGT is divided into two stages. The first stage is the increasing of the lower heating value of the original ammonia by the transformation of exhaust heat into the chemical energy of hydrogen-rich fuel. In this stage, a new hydrogen-rich fuel with combustion heat higher than combustion heat of initial ammonia (in comparison to 1 kg of initial ammonia) is obtained. And the second stage is combustion of this new hydrogen-rich fuel in a combustion chamber [30].

In the reformer the endothermic reaction of ammonia decomposition is occurring. This reaction is well studied [31,32]. The chemical reaction of ammonia decomposition is as follows:



2.2. Thermodynamic analysis

The thermodynamic model has been developed and realized via Aspen HYSYS which is a powerful process simulation tool for schematic design of various concepts. Based on the developed model, parametric analysis was performed for ammonia-fired CRGT and ammonia fired without exhaust heat recuperation. The main goal of this analysis was to determine which operating parameters have a significant effect on overall CRGT efficiency and to determine the operating parameters for maximum CRGT efficiency. There is a large number of publications on using Aspen HYSYS for modeling of various schemes of gas turbine units [33–35].

The calculation scheme in Aspen HYSYS for ammonia-fired CRGT is presented in Fig. 2. The temperature of working fluid at the gas turbine inlet has been regulated by the air mass flow rate. Aspen HYSYS has a useful function, (*Adjust*), in which the user sets the target variable value and specifies an adjusted variable used to reach the target value. The temperature of exhaust gas has been set

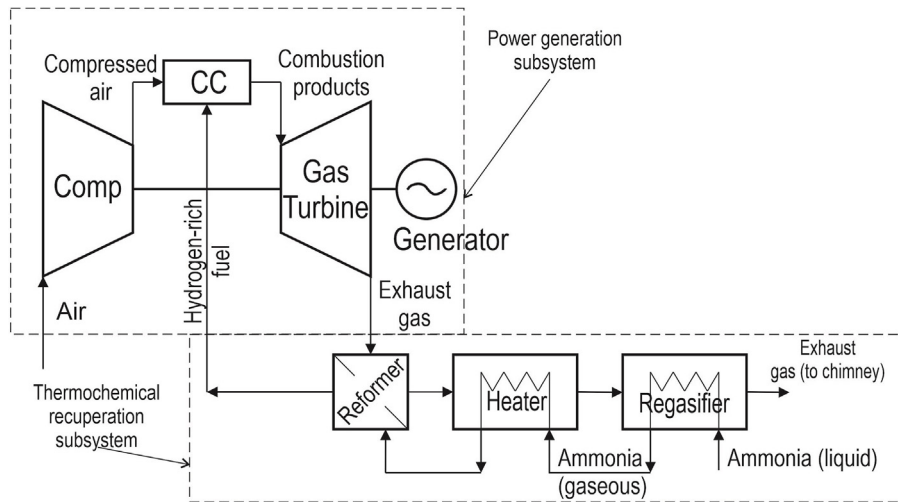


Fig. 1. The schematic diagram of an ammonia-fired chemical recuperated gas turbine.

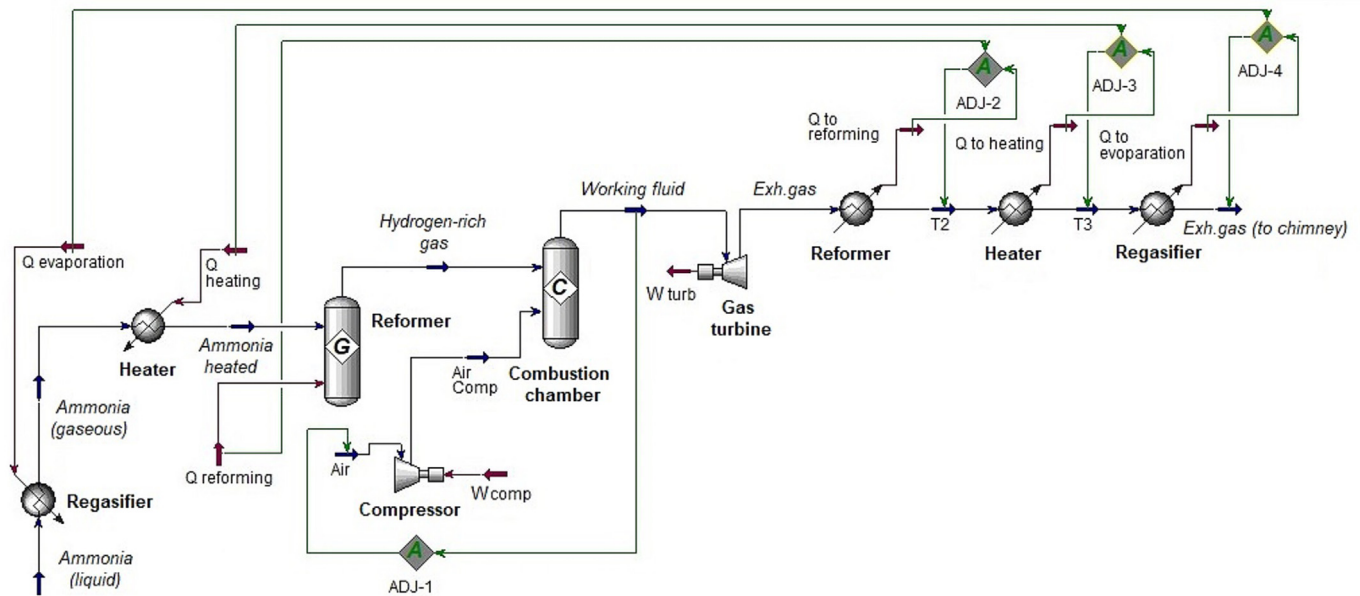


Fig. 2. The computational schemes of ammonia fired CRGT in Aspen HYSYS.

as the target variable and the air mass flow as the adjusted variable. Also, we used *Adjust* to compile the heat balance for the determination of exhaust gas temperature after a reformer, a heater, and a regasifier.

In this paper, the net power output of the gas turbine has been calculated based on the following expression:

$$W_{net} = W_{gt} - W_{comp} \quad (2)$$

where W_{gt} – the power produced in gas turbine; W_{comp} – the power for air compressing (compressor work).

In order to make a good comparative analysis, the working fluid temperature at the gas turbine inlet has been set as constant for all analyzed schemes (CRGT and gas turbine without recuperation). The temperature range of working fluid at the gas turbine inlet is from 700 to 1300 °C. Other initial conditions and turbine parameters for a parametric analysis are presented in Table 1.

The recuperated heat for CRGT is a sum of heat that has been transferred from exhaust gas to ammonia in the reformer for the decomposition process and the ammonia heater as well as heat for ammonia evaporation. In other words, the recuperated heat in CRGT can be determined as a sum of the enthalpy of the ammonia

Table 1

Stream properties and efficiency parameters of gas turbine and compressor.

Parameter	Value
Ammonia mass flow rate	100 kg/h
Turbine isentropic efficiency	90%
Compressor isentropic efficiency	85%
Decomposition temperature	$T_{exh} - 50$ °C
Inlet temperature of ammonia	20 °C
Temperature of atmospheric air	20 °C
Pressure of all inlet streams (except atmospheric air)	6–21 bar

decomposition reaction, the enthalpy change of ammonia in a heater, and the enthalpy of a regasification process. In the CRGT the recuperated heat can be determined as follows:

$$H_{rec} = \Delta H_{298} + \Delta H_{heat} + \Delta H_{regas} \quad (3)$$

where ΔH_{298} – the enthalpy of the ammonia decomposition reaction; ΔH_{heat} – the enthalpy change of ammonia in a heater; ΔH_{sg} – the enthalpy of a regasification process.

One of the main criteria of the efficiency of the thermochemical recuperation systems is a heat recovery rate (HRR). HRR is a relation between the recuperated heat and the enthalpy of exhaust heat after a gas turbine. Therefore, the heat recuperation rate can be determined via the following expression:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{\Delta H_{298} + \Delta H_{heat} + \Delta H_{regas}}{H_{exh}} \quad (4)$$

where H_{exh} – an exhaust gas enthalpy.

Moreover, in this paper, the recuperated heat in a reformer, a heater, a regasifier as a share of the total recuperated heat in TCR system was determined from the following expression:

$$r_{rec,i} = \frac{H_{rec,i}}{H_{rec,total}} \quad (5)$$

where $H_{rec,i}$ – the recuperated heat in the i th element of the scheme (reformer, heater, regasifier); $H_{rec,total}$ – the total recuperated heat in whole TCR system.

The enthalpy of the reaction of ammonia decomposition in the expression (4) is depending on the operational parameters such as temperature and pressure. Therefore, the thermodynamic analysis includes not only the cycle thermodynamics but also the thermodynamic of the ammonia decomposition process.

To determine the enthalpy of the decomposition reaction and hydrogen-rich gas composition as a function of the operational parameters, the thermodynamic analysis via Gibbs free energy minimization method was performed.

The thermodynamic analysis makes it possible the calculation of the equilibrium concentration of the products of ammonia decomposition at the given technological parameters such as temperature and pressure. The main concept of the Gibbs free energy minimization method is the fact the reaction system is thermodynamically favorable when its total Gibbs free energy is minimum and its differential is equal to zero [36,37]:

$$dG^t = 0 \quad (6)$$

where G^t – total Gibbs free energy.

Total Gibbs free energy of the ammonia decomposition process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \quad (7)$$

where n_i – number of i th component moles; μ_i – chemical potential of i th component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (8)$$

where G_{fi}^0 – standard Gibbs free energy of i th component formation; R – molar gas constant; T – system temperature; f_i – fugacity

of a pure i th component; f_i^0 – fugacity of a pure i th component at the standard state.

If we combine the equations presented above, we can write the equation for total Gibbs free energy of the chemical system:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (9)$$

The thermodynamic analysis of TCR system was also performed via Aspen-HYSYS according to the calculation scheme from Fig. 2. To simulate the ammonia decomposition process, the ideal Gibbs reactor is chosen. The Gibbs reactor determines the decomposition products by the Gibbs free energy minimization method of each component based on the chemical mechanism of the ammonia decomposition process. The Peng-Robinson equation of state was used to calculate the volume of the product gas as a function of pressure and temperature [38].

The thermodynamic analysis doesn't consider the chemical kinetics, effect of catalyst, residence time, etc. on the decomposition products. The thermodynamic analysis gives us opportunity to determine the maximum (ultimate level) of the energy efficiency for ammonia fired CRGT at given operation parameters such as temperature and pressure. These limit levels of the energy efficiency can be a guideline for the design of the elements of the thermochemical recuperation systems by ammonia fired CRGT.

3. Results and discussion

3.1. Equilibrium conditions

As mentioned above the thermodynamic calculations were performed with the help of Aspen HYSYS. This software is a recognized simulator for modeling the chemical process. However, in this paper, we performed verification of results to show that our results are accurate. The results of the thermodynamic analysis were compared with the results that calculated from experimental data of Fixed Nitrogen Research Laboratory (F.N.R.L.) [39]. Fig. 3 shows the comparison of the results obtained in this paper with the calculated results from experimental data of Fixed Nitrogen Research Laboratory (F.N.R.L.) [39].

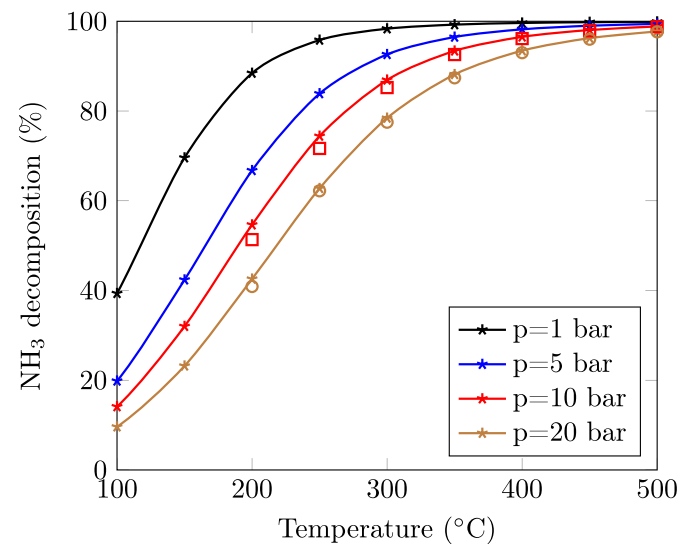


Fig. 3. Comparison of the results of thermodynamic analysis (solid lines) with the calculated from experimental data of Fixed Nitrogen Research Laboratory (F.N.R.L.) [39] (hollow points).

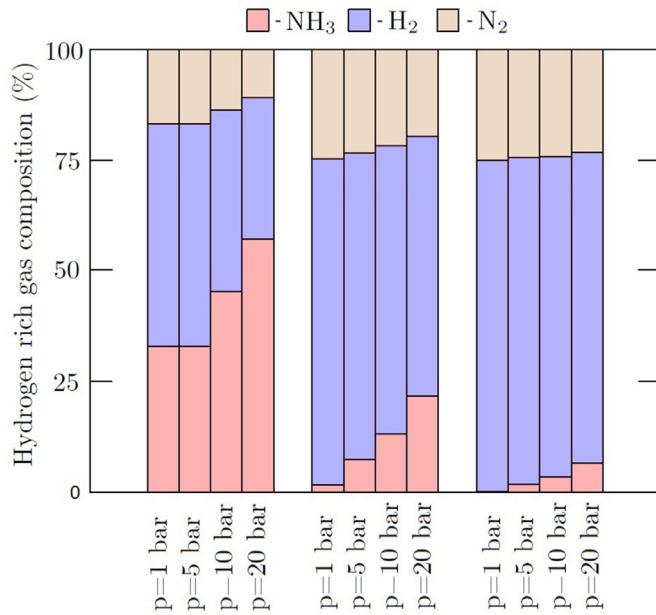


Fig. 4. The hydrogen rich gas composition after ammonia decomposition.

Fig. 3 shows that our results of thermodynamic analysis and the results that calculated from experimental data of Fixed Nitrogen Research Laboratory (F.N.R.L.) [39] are in a good correlation. The results for pressure of 20 bar for verification in Fig. 3 were extrapolated from the data presented in Table V in this ref. [39].

The equilibrium composition of hydrogen-rich gas after the reactor depends on temperature and pressure. Previously we determined the ammonia gas composition [40]. Fig. 4 presents the hydrogen-rich gas composition for various operating parameters. Fig. 4 makes it clear that product composition strongly depends on pressure and temperature. While at 200 °C H_2 concentration is less than 50%, at 400 °C about three-quarters of the volume is hydrogen. With regard to the problem of thermochemical waste-heat recuperation as on-board hydrogen production technology, this fact must be taken into account. Consequently, to achieve the maximum hydrogen yield in the TCR systems, it is necessary to have a heat source (for example, exhaust flue gases) with a temperature above 300–400 °C. Fig. 4 shows that increasing temperature leads to increasing in hydrogen yield as well as decreasing in remain ammonia in the hydrogen-rich gas. Pressure also is playing an important role in the ammonia decomposition process. For example, at a temperature of 300 °C for 1 bar the share of ammonia in the product gas is about 2% while at the same temperature for 20 bar the share of ammonia is about 20%. Therefore, pressure also has to be taken into account for the application of ammonia decomposition in the thermochemical waste-heat recuperation systems.

The efficiency of the thermochemical waste-heat recuperation according to Eq. (4) depends on the enthalpy of the reaction decomposition. Fig. 5 shows the effect of the operating parameters on the enthalpy of the reaction decomposition.

Fig. 5 depicts that temperature increasing leads to an increase in the reaction enthalpy. In a temperature range from 100 to 300 °C the effect of temperature is especially noticeable. In a temperature range 300–400 °C the reaction enthalpy is slightly increasing and in the temperature range above 400 °C is reaching its maximum values. On the other hand, the pressure increasing leads to a decrease in the reaction enthalpy. For temperature 250 °C the reaction enthalpy at 1 bar is approximately twice higher than the

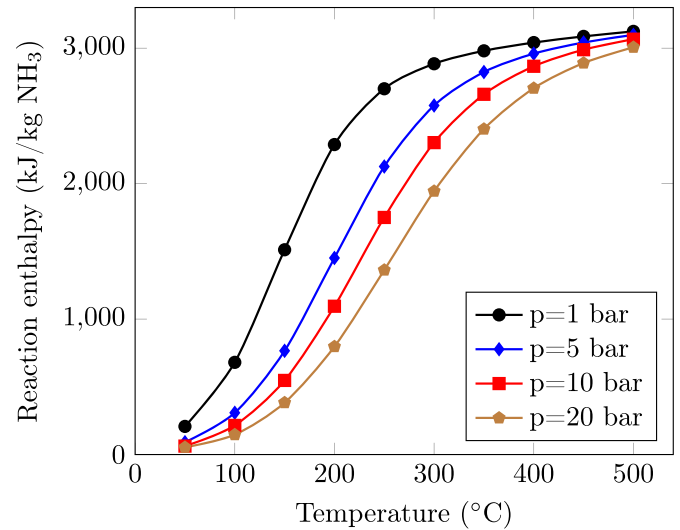


Fig. 5. The enthalpy of the ammonia decomposition reaction.

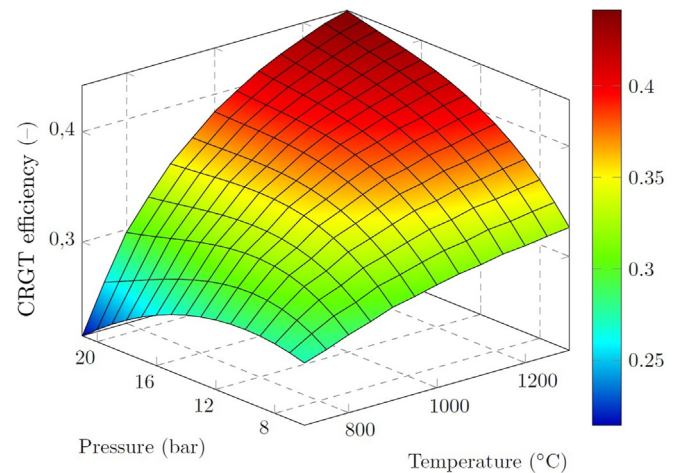


Fig. 6. Efficiency of CRGT vs. inlet temperature and pressure.

reaction enthalpy at 20 bar. Therefore, an increase in ammonia pressure leads to a decrease in storage volume. However, on the other hand, it leads to a decrease in the reaction enthalpy.

3.2. Gas turbine efficiency

Exhaust heat recuperation leads to an increase in the efficiency of the gas turbine. To determine the efficiency factor of CRGT, the following expression is used:

$$\eta_{gtu} = \frac{W_{gt} - W_{comp}}{m_{NH_3} \cdot LHV_{NH_3}} = \frac{W_{net}}{m_{NH_3} \cdot LHV_{NH_3}} \quad (10)$$

where W_{comp} is the compressor power, W_{gt} is the gas turbine power, m_{NH_3} is the ammonia mass flow; and LHV_{NH_3} is the ammonia lower heating value.

Fig. 6 shows the efficiency of CRGT as a function of inlet temperature (700–1300 °C) and pressure (6–21 bar). Fig. 6 makes it clear that an increase in pressure in the temperature range up to 900 °C leads first to an increase in efficiency, and after reaching the maximum value, leads to a decrease. In this case, the decrease in efficiency is due to the fact that the increase in compressor

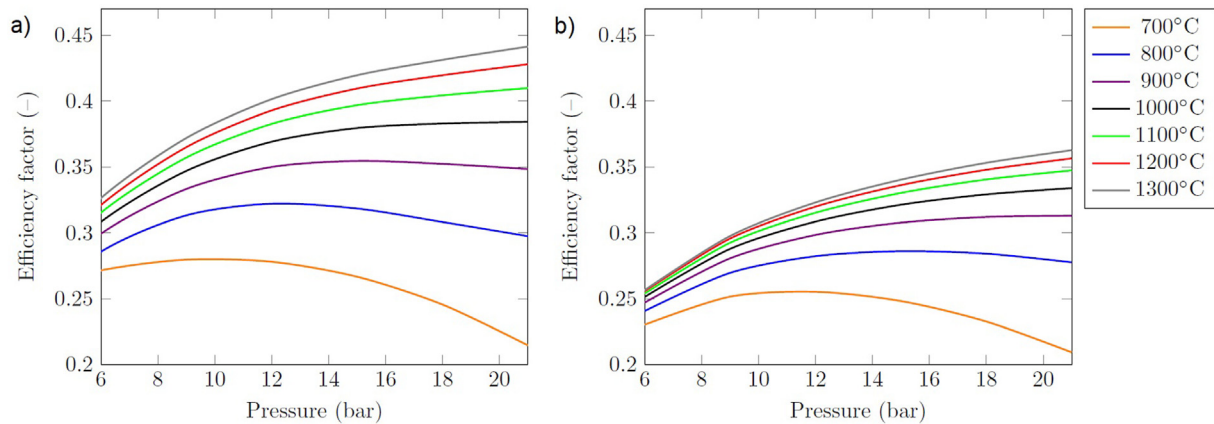


Fig. 7. The comparison of the efficiency of CRGT (a) and gas turbine without recuperation (b).

operation is faster than the increase in gas turbine power. In the temperature range above 900 °C the efficiency increases with an increase in pressure and temperature. However, the maximum increase in efficiency occurs in the pressure range up to 15 bar. When the pressure rises above 15 bar, the efficiency increase insignificantly. The results of calculating the CRGT scheme in Aspen HYSYS are attached as supplementary materials to this paper.

To compare the efficiency of CRGT with a gas turbine without a recuperation system, we presented Fig. 7. Fig. 7 shows the comparison of the efficiency of CRGT and gas turbine without recuperation. Fig. 7 depicts that the addition of thermochemical exhaust heat recuperation leads to a significant increase in efficiency, especially in the temperature range above 800 °C. The maximum efficiency of CRGT is observed at 9 bar for $T_{in} = 700$ °C; 12 bar at $T_{in} = 800$ °C; 15 bar at $T_{in} = 900$ °C; 18 bar at $T_{in} = 1000$ °C. While the maximum efficiency of gas turbine without recuperation is observed at 11 bar at $T_{in} = 700$ °C; 16 bar at $T_{in} = 800$ °C, 20 bar at $T_{in} = 900$ °C.

The effect of pressure for each inlet temperature on the CRGT efficiency is shown in Fig. 8. Fig. 8 shows at low pressures, an increase in temperature leads to a slight increase in efficiency. However, as the pressure rises, the increase in efficiency increases. For example, for a pressure of 21 bar, the efficiency more than doubles with an increase in temperature from 700 to 1300 °C.

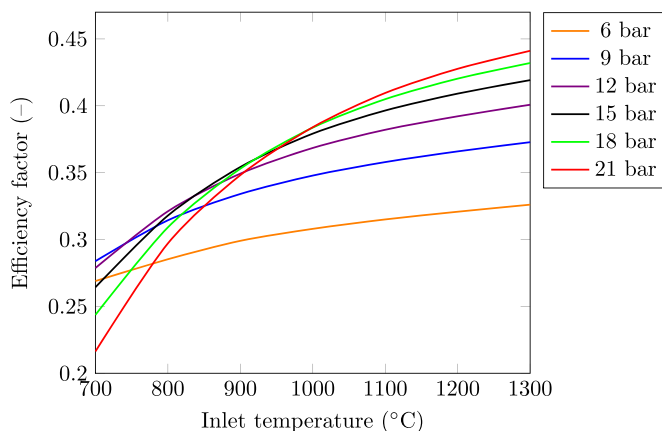


Fig. 8. Effect of pressure for each inlet temperature on the CRGT efficiency.

At a temperature of 960 °C, the CRGT efficiency for pressures of 15, 18, and 21 bar is the same and equals 0.37. Fig. 8 shows that maximum CRGT efficiency at moderately low inlet temperatures can be obtained at reduced pressure. In the temperature range of 800–1000 °C the optimal pressure for maximization of CRGT efficiency at moderately low pressure is 15 bar.

Some results of the thermodynamic analysis are presented in Table 2. In this table, the work of compressor (W_{comp}), the power of gas turbine (W_{gt}), the net power, the air mass flow rate (G_{air}), and CRGT efficiency (η_{gtu}) is plotted. The complete results of the thermodynamic analysis are added to this paper as supplementary materials.

3.3. Exhaust heat recuperation

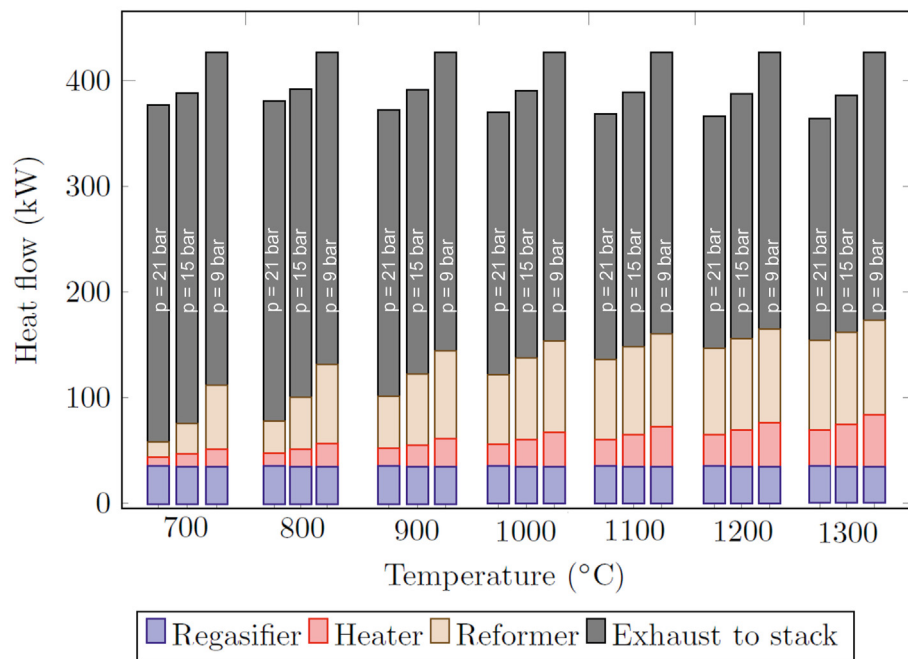
The recuperated heat for CRGT is a sum of heat that has been transferred from exhaust gas to ammonia in the reformer for the decomposition process and the ammonia heater as well as heat for ammonia evaporation. In other words, the recuperated heat in CRGT can be determined as a sum of the enthalpy of the ammonia decomposition reaction, the enthalpy change of ammonia in a heater, and the enthalpy of a regasification process. All obtained results for exhaust heat recuperation can be found in the supplementary materials. Below we give a discussion of some interesting facts.

Fig. 9 shows the heat flow of the exhaust gases after the gas turbine. The whole heat flow of the exhaust gases is spent for ammonia regasification, ammonia preheating, ammonia decomposition, and the exhaust heat to stack. Fig. 9 presents the results for pressure of 9, 15, 21 bar. The total column height is the heat of the exhaust gases. From Fig. 9 it can be seen that the heat flow for ammonia regasification is almost constant for all operating parameters while the heat flow for ammonia decomposition is depending on pressure and temperature especially in the temperature range below 1000 °C. This fact is in good agreement with the results presented in Fig. 5. An increase in temperature leads to an increase in the enthalpy of the decomposition process; however, in the temperature range above 500 °C, the enthalpy tends to a maximum value. Moreover, an increase in pressure decreases the enthalpy of the decomposition reaction. Fig. 9 shows that the distribution of recuperated heat in each element of the recuperation system (regasifier, heater, reformer) is different for various operating parameters.

Table 2

The power of compressor, the power of gas turbine, the net power, the air mass flow rate and the CRGT efficiency.

Inlet temperature, °C	Pressure, bar	W_{comp} , kW	W_{gt} , kW	W_{net} , kW	G_{air} , kg/h	η_{gtu}
1200	6	118.9	319.4	200.5	1846	0.321
	9	163.2	391.9	228.7	1946	0.366
	12	201.6	446.7	245.1	2029	0.392
	15	236.4	492.1	255.7	2110	0.409
	18	268.9	531.6	262.7	2186	0.420
	21	299.7	567.0	267.3	2257	0.428
1000	6	153.2	345.7	192.5	2379	0.308
	9	214.5	431.9	217.4	2554	0.348
	12	269.1	499.4	230.3	2710	0.368
	15	319.5	556.5	237.0	2852	0.379
	18	367.0	606.8	239.8	2983	0.384
	21	412.4	652.4	240.0	3106	0.385
800	6	211.7	390.0	178.3	3287	0.285
	9	304.2	500.7	196.5	3621	0.314
	12	388.8	589.5	200.7	3915	0.321
	15	470.1	668.8	198.7	4196	0.318
	18	552.7	745.9	193.2	4492	0.309
	21	639.7	825.4	185.7	4818	0.297

**Fig. 9.** Effect of pressure for each inlet temperature on the CRGT efficiency.

The recuperated heat in a regasifier, a heater, a reformer as a share of the total recuperated heat in the thermochemical recuperation system at the pressure of 9, 15, 21 bar is presented in Fig. 10.

Fig. 10 shows that the share of the recuperated heat in a reformer is increasing in the temperature range below 1000 °C with temperature increasing. At the same time, the share of the recuperated heat in a regasifier is decreasing when temperature increases. This is due to the fact that the recuperated heat in a gasifier is constant when the recuperated heat in a heater and a reformer is increasing. Moreover, Fig. 10 shows that in the temperature range above 1000 °C the share of recuperated heat in each element of the TCR system is slightly changing. About half of the total recuperated heat is recovered in the reformer and the second half of recuperated heat is recovered in the heater and regasifier.

To determine the efficiency of the thermochemical exhaust heat

recuperation system, the heat recovery rate was determined. Fig. 11 show the heat recovery rate as a function of pressure for different temperatures. The sharp increase in HRR in the pressure range from 6 to 9 bar is due to the fact that at a pressure of 6 bar, ammonia is in a gaseous state. Therefore, the recuperated heat in the regasifier is zero. A further increase in pressure will lead to a decrease in HRR because the denominator in expression (4) increases. At temperatures above 1200 °C, an increase in pressure does not lead to a noticeable change in HRR.

The thermochemical exhaust heat recuperation system makes it possible to recover up to 43% of exhaust heat. Therefore, there is a potential to use other recuperation systems for improvement in the energy efficiency of ammonia fire CRGT, for example, steam cycle or steam injection. To determine such potential, the temperature of the exhaust gases after the gas turbine and the temperature of the exhaust gases after the thermochemical recuperation system

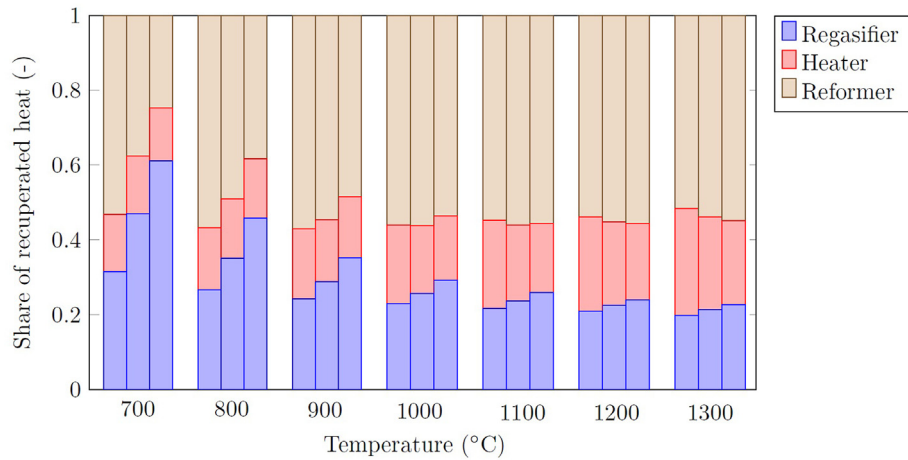


Fig. 10. Recuperated heat in a regasifier, a heater, a reformer as a share of the total recuperated heat in the TCR system at the pressure of 9, 15, 21 bar.

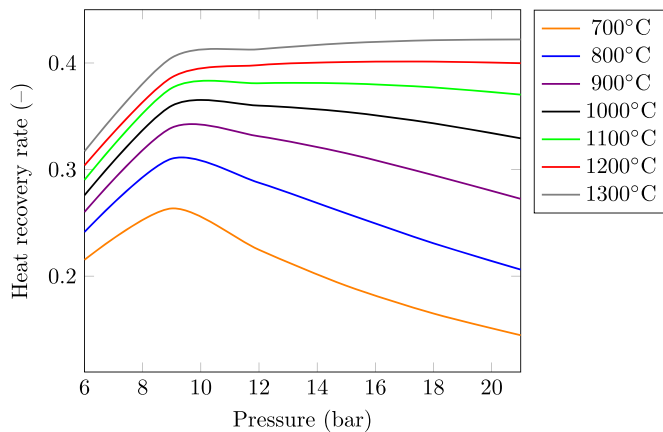


Fig. 11. Heat recovery rate as a function of pressure for different temperatures.

(exhaust gas to stack) were compared for various operating parameters. Fig. 12 shows the temperature of the exhaust gases after the gas turbine and temperature of the exhaust gases after the thermochemical recuperation system (exhaust gas to stack) as a function of inlet temperature (from 700 to 1300 °C) and pressure.

Fig. 12 shows that the temperature of the exhaust gases after the TCR systems have a big potential for the addition of other recuperation systems.

4. Conclusions

The thermodynamic analysis of the cycle and recuperation system of an ammonia-fired chemically recuperated gas turbine (CRGT) is performed. The thermochemical exhaust heat recuperation system is based on the endothermic reaction of ammonia decomposition. The analysis is performed for a wide range of operating parameters: turbine inlet temperature of 700–1300 °C, the pressure of 6–21 bar. The comparison of the efficiency of CRGT and gas turbine without recuperation depicts that the addition of the thermochemical exhaust heat recuperation system leads to a significant increase in the efficiency, especially in the temperature range above 800 °C.

The maximum efficiency of ammonia-fired CRGT is observed for the following conditions: 9 bar for $T_{in} = 700$ °C; 12 bar at $T_{in} = 800$ °C; 15 bar at $T_{in} = 900$ °C; 18 bar at $T_{in} = 1000$ °C. While the maximum efficiency of gas turbine without recuperation is observed at 11 bar at $T_{in} = 700$ °C; 16 bar at $T_{in} = 800$ °C, 20 bar at $T_{in} = 900$ °C. The maximum CRGT efficiency for the analyzed operating parameters is 0.44 for ($T_{in} = 1300$ °C and $p = 21$ bar).

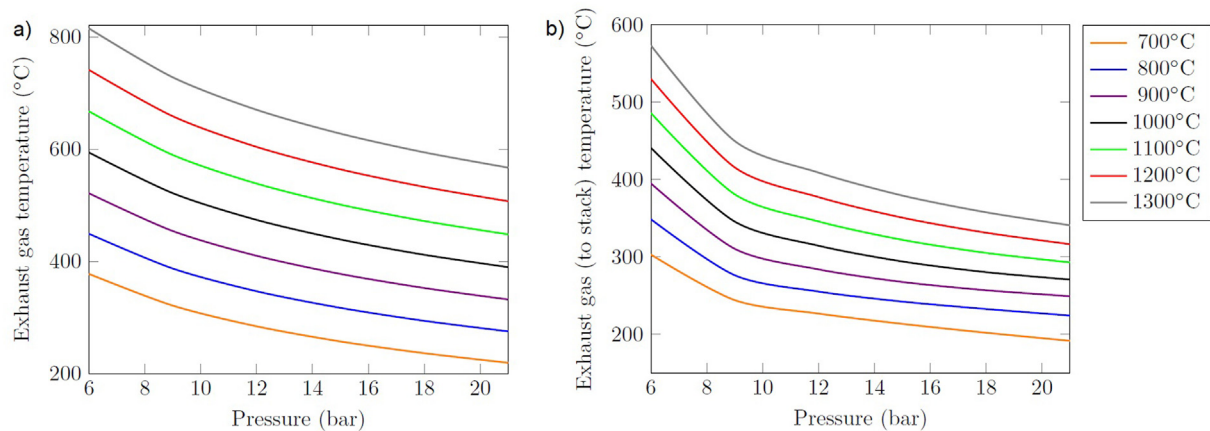


Fig. 12. Temperature of the exhaust gases after gas turbine and temperature of the exhaust gases after the thermochemical recuperation system (exhaust gas to stack) as a function of inlet temperature and pressure.

The heat balance of the recuperation system showed that about half of the total recuperated heat is recovered in the reformer and the second half of recuperated heat is recovered in the heater and regasifier. The thermochemical exhaust heat recuperation system makes it possible to recover up to 43% of exhaust heat. Therefore, there is a potential to use other recuperation systems for improvement in the energy efficiency of ammonia fire CRGT, for example, steam cycle or steam injection. The temperature of the exhaust gases after the recuperation systems has a big potential for the addition of other recuperation systems because the temperature of the exhaust gas after the recuperation system is 300–450 °C for the inlet temperature of 1000–1300 °C.

Credit author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2022.124081>.

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